164994 - Ndinya Bright Seya

Part A

1. a) $R = 20 + 30 + 50$

$= 100\,\Omega$

b) $R = 1/20 + 1/100 + 1/50$

$= 2/25 = 0.08\,\Omega$

2. a) $I = V/R$

$= 125V/100\,\Omega$

$= 1.25A$

b) $I = V/R$

$= 125V/0.08\,\Omega$

$= 1562.5A$

3. a) $I_{20\Omega} = I_{30\Omega} = I_{50\Omega} = I_T$

$I_T = 1.25A$

$I_{20\Omega} = 1.25A$ $I_{30\Omega} = 1.25A$ $I_{50\Omega} = 1.25A$

b) $I_n = V/R_n$

$I_{20\Omega} = 125V/20\,\Omega$

$= 6.25A$

$I_{100\Omega} = 125V/100\,\Omega$

$= 1.25A$

$I_{50\Omega} = 125V/50\,\Omega$

$= 2.5A$

OnlinePhotoScanner.com

4. a) $V_n = IR_n$

$V_{20\Omega} = 1.25A \times 20\Omega$
$= 25V$

$V_{50\Omega} = 1.25A \times 50\Omega$
$= 62.5V$

$V_{30\Omega} = 1.25A \times 30\Omega$
$= 37.5V$

$V_{30\Omega} - V_{20\Omega} = 12.5V$

$V_{50\Omega} - V_{30\Omega} = 25V$

b) $V_1 = V_2 = V_3 = V_T$

$\therefore V_{20\Omega} = IR = 6.25A \times 20\Omega$
$= 125V$

$V_{100\Omega} = 1.25A \times 100\Omega$
$= 125V$

$V_{50\Omega} = 2.5A \times 50\Omega$
$= 125V$

$\therefore$ 0 voltage drop
since $V_{20\Omega} = V_{100\Omega} = V_{50\Omega}$

5. a) $P_{20\Omega} = 25V \times 1.25A$
$= 31.25W$

$P_{50\Omega} = 62.5V \times 1.25A$
$= 78.125W$

$P_{30\Omega} = 37.5V \times 1.25A$
$= 46.875W$

b) $P_{20\Omega} = 125V \times 6.25A$
$= 781.25W$

$P_{100\Omega} = 125V \times 1.25A$
$= 156.25W$

$P_{50\Omega} = 125V \times 2.5A$
$= 312.5W$

OnlinePhotoScanner.com